

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
4 HT	(a)	(i)		Aluminium is a better or good conductor (1) The metal spacer conducts heat from the inner <u>pane</u> to the outer <u>pane</u> or from one <u>pane</u> to the <u>other</u> (1) <u>Increased vibrations</u> pass on energy from one particle to another (1) Also <u>free (or delocalised) electrons</u> transfer energy [from high T to low T] (1) Alternative: uPVC is <u>a worse or poor conductor</u> or a good insulator (1) The non-metal spacer conducts heat poorly from the inner <u>pane</u> to the outer <u>pane</u> or from one <u>pane</u> to the <u>other</u> (1) <u>Increased vibrations</u> pass on energy from one particle to another [less effectively] (1) <u>No free (or delocalised) electrons</u> to transfer energy [from high T to low T] (1)	1 1	 1 1		4		
		(ii)		Heat from the house <u>conducts</u> through the <u>ceiling</u> into the attic or heat <u>conducts</u> through the <u>roof tiles</u> (1) <u>Air</u> is heated it becomes <u>less dense and rises</u> (1) Creating a convection [current] in the <u>roof space</u> / <u>outside air</u> becomes heated by the roof tiles causing a convection current (1)	3			3		
	(b)			Energy needed to maintain the temperature / heating is used to replace energy lost / energy in = energy out (1) If you lose less heat your heating doesn't need to be on for as long (1) So less fuel is burned / less use of fossil fuels / less mining required / less transport of fuels / fewer power stations / less CO ₂ or SO ₂ [emissions] / less [contribution to] global warming or greenhouse effect / so less [contribution to] acid rain (1) Don't accept reference to pollution	3			3		
				Question 4 total	8	2	0	10	0	0